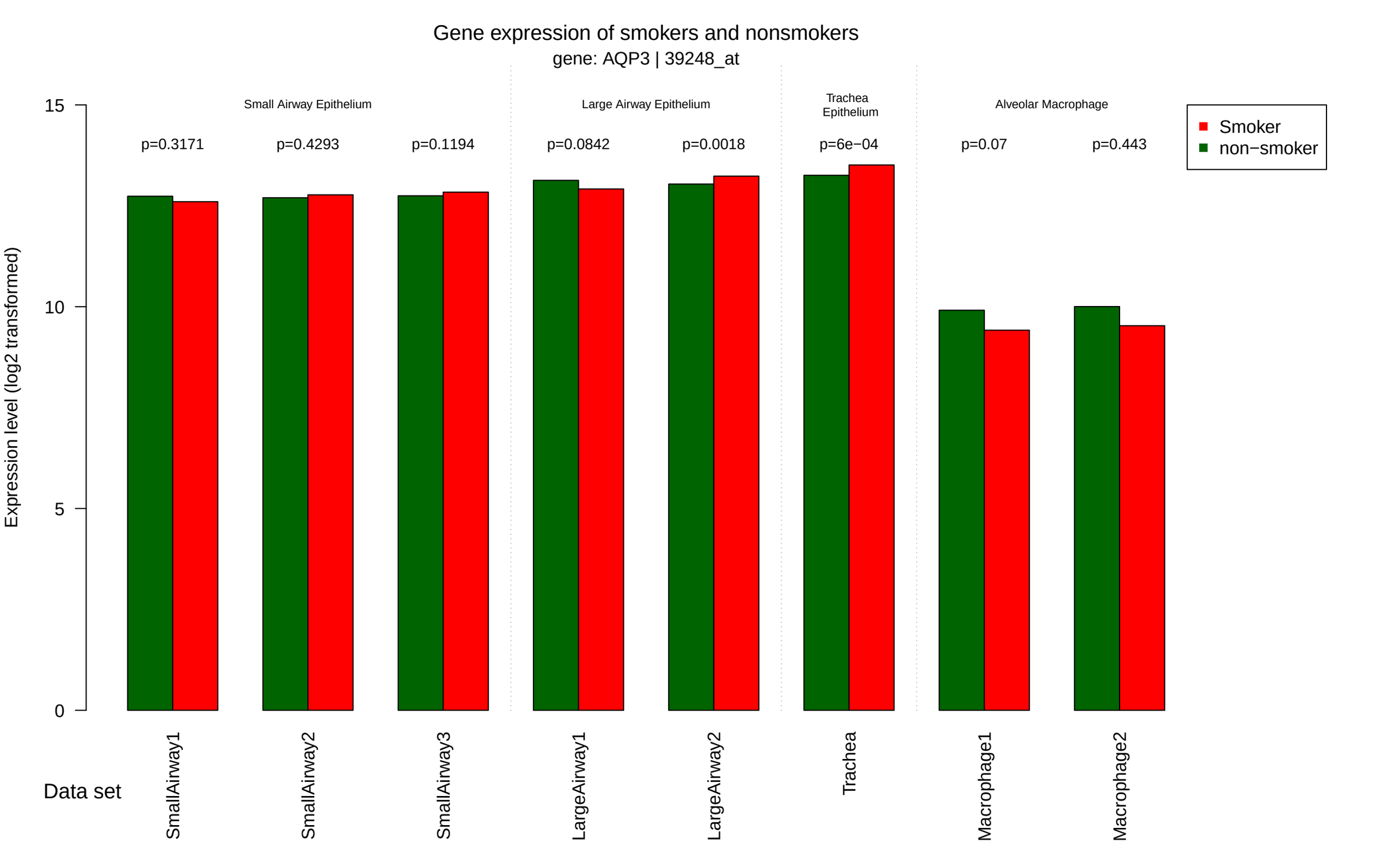
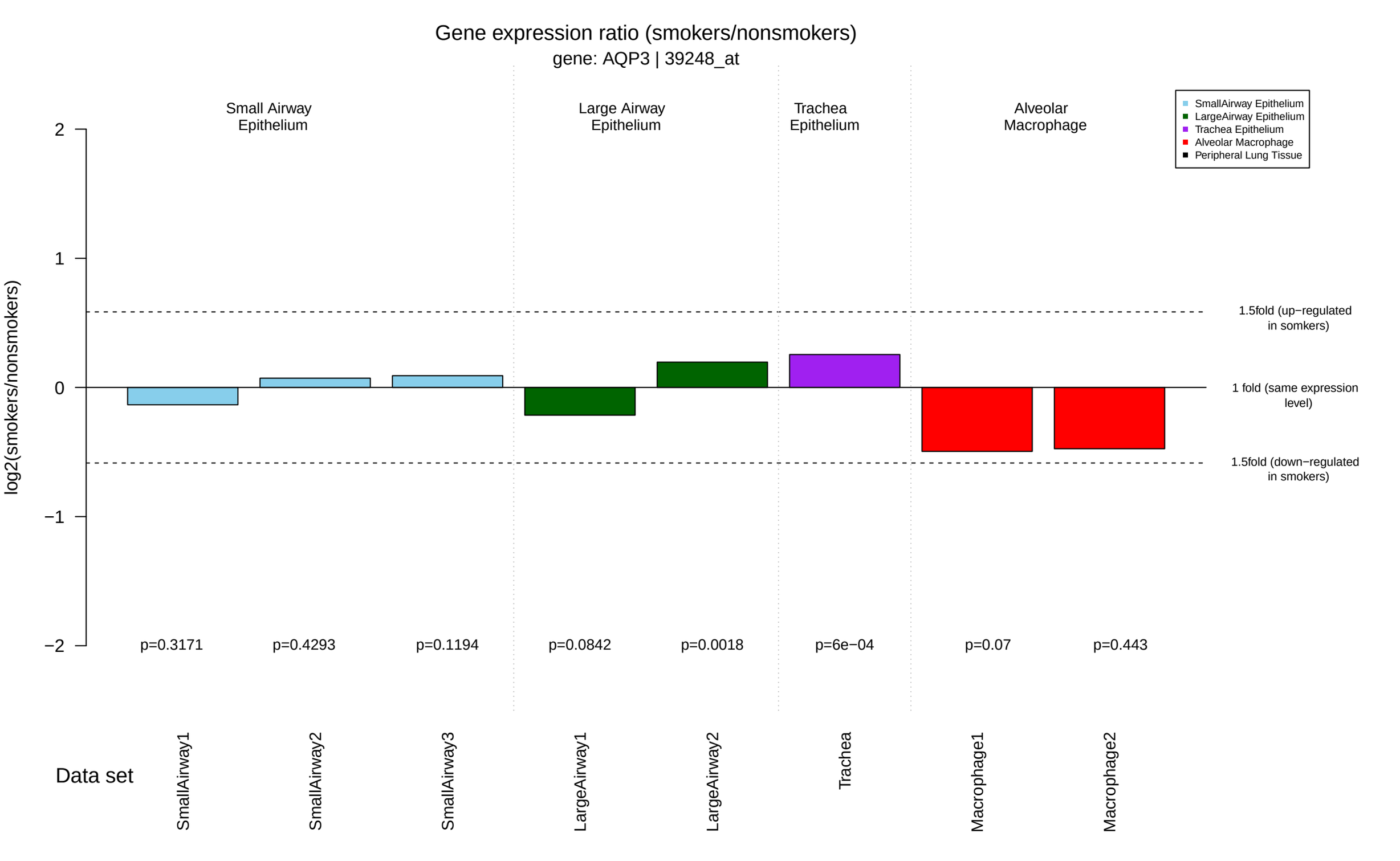
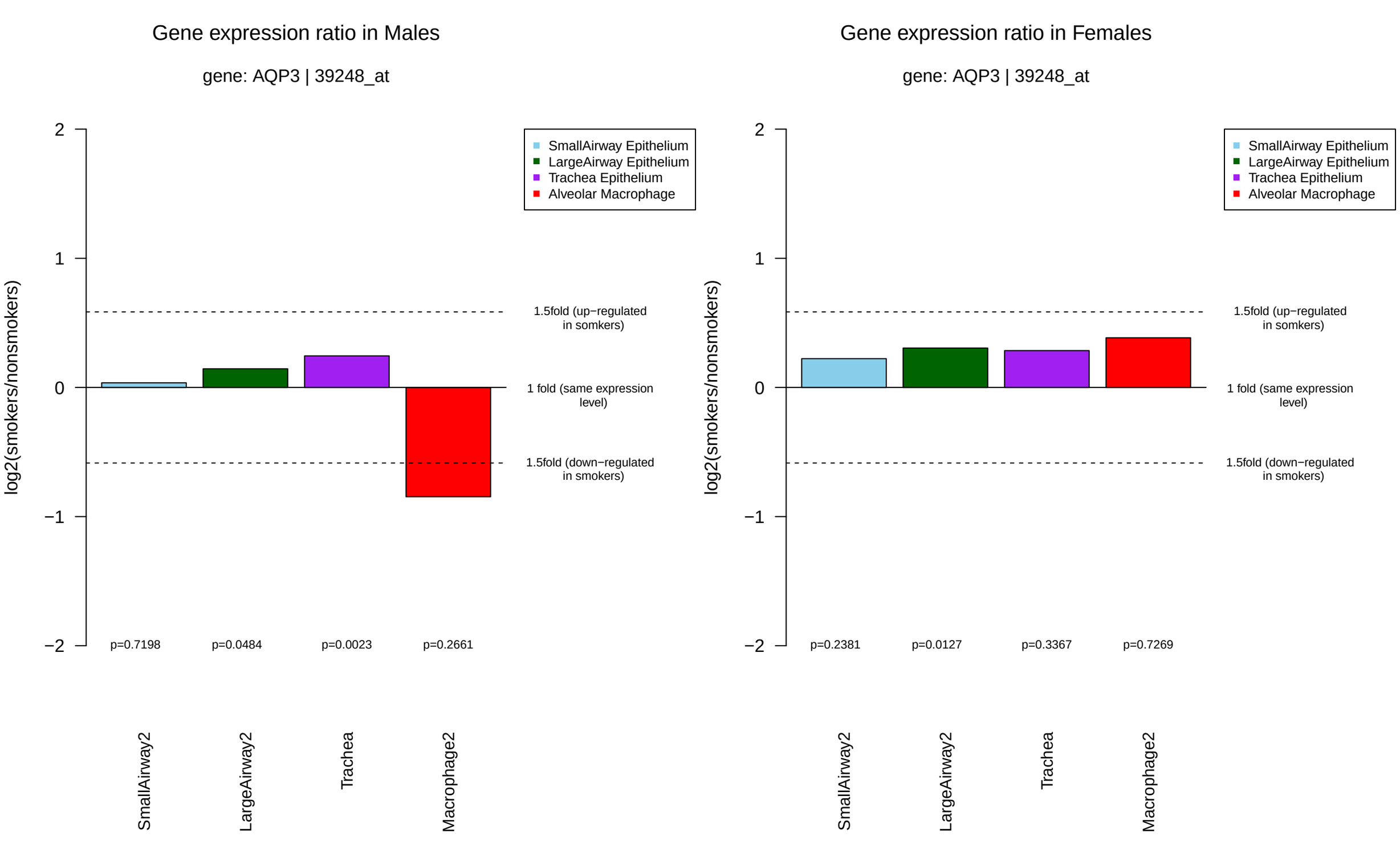
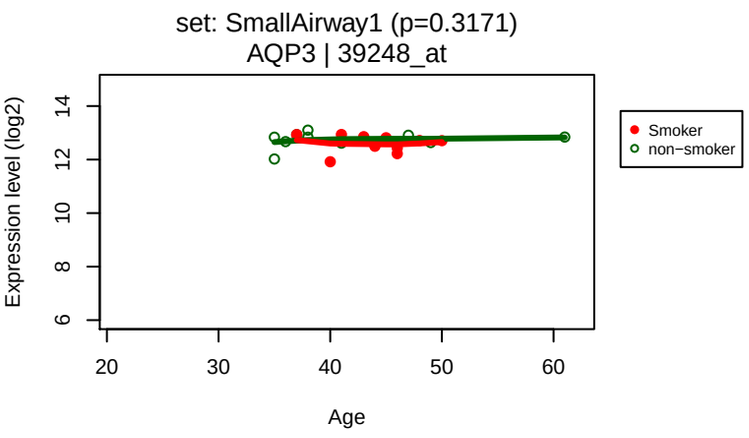
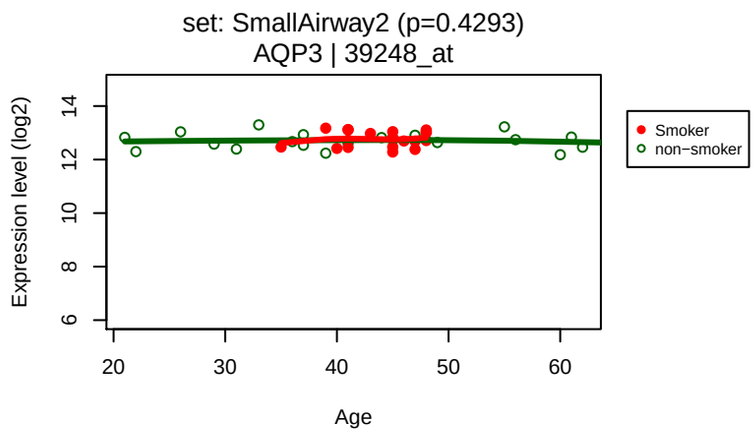
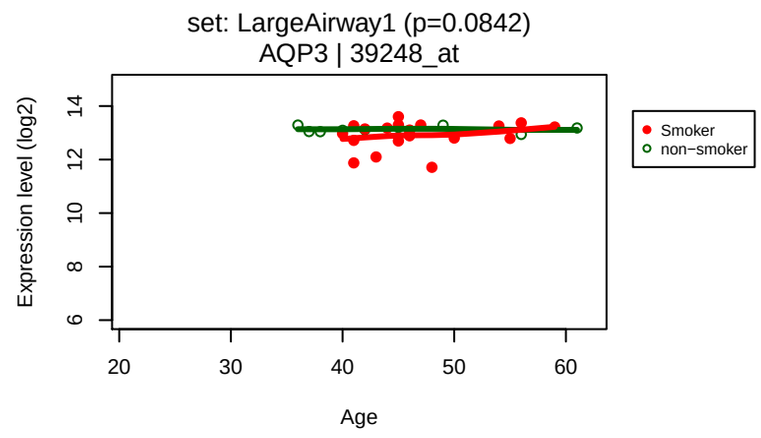

Gene expression of smokers and nonsmokers
gene: AQP3 | 39248_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3171
p=0.4293
p=0.1194
p=0.0842
p=0.0018
p=6e−04
p=0.07
p=0.443
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

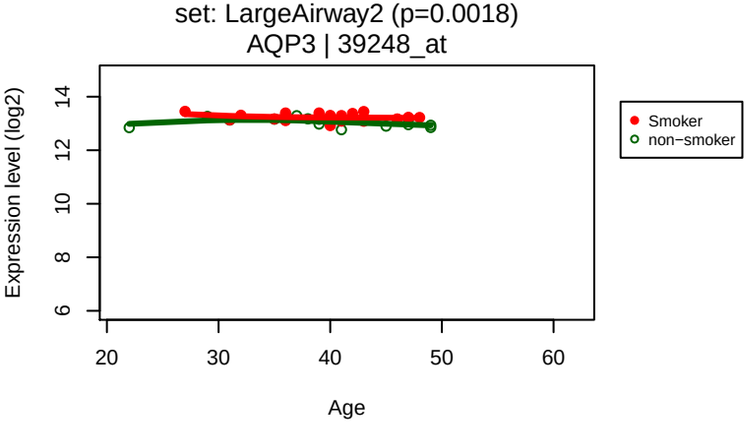

Gene expression ratio (smokers/nonsmokers)
gene: AQP3 | 39248_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3171
p=0.4293
p=0.1194
p=0.0842
p=0.0018
p=6e−04
p=0.07
p=0.443
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

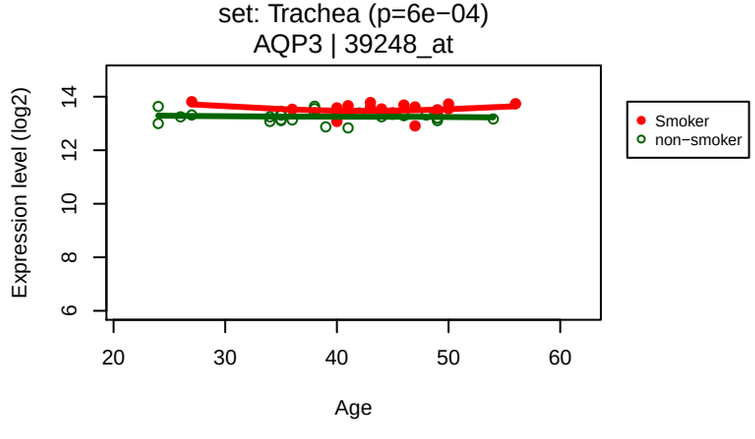

Gene expression ratio in Males
gene: AQP3 | 39248_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7198
p=0.0484
p=0.0023
p=0.2661
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AQP3 | 39248_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2381
p=0.0127
p=0.3367
p=0.7269
SmallAirway2
LargeAirway2
Trachea
Macrophage2

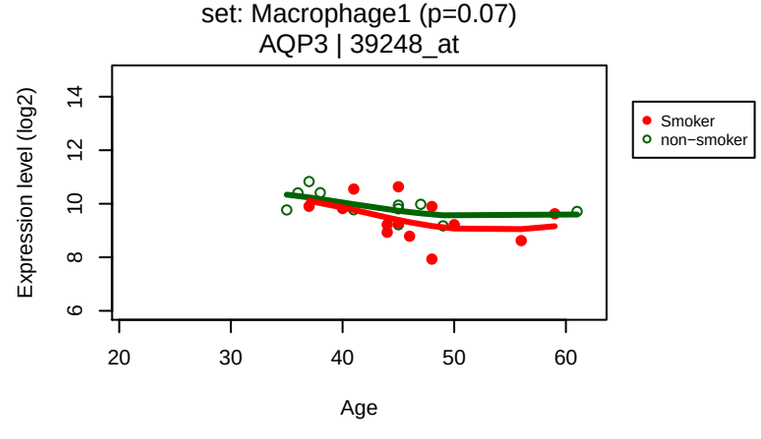

set: SmallAirway1 (p=0.3171)
AQP3 | 39248_at
Expression level (log2)
Age
Smoker
non−smoker

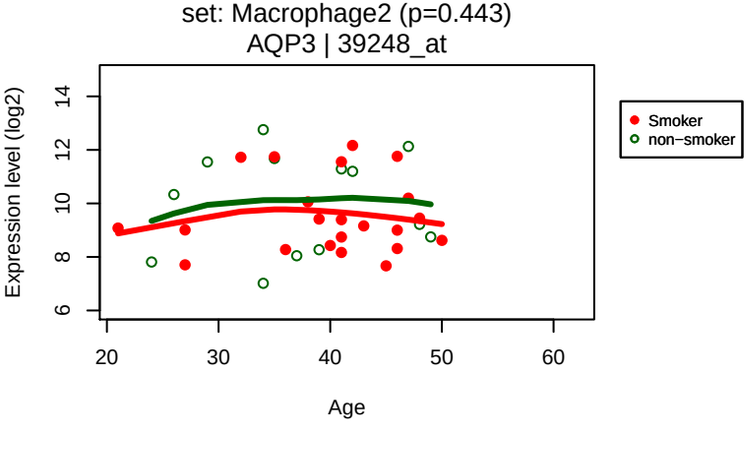

set: SmallAirway2 (p=0.4293)
AQP3 | 39248_at
Expression level (log2)
Age
Smoker
non−smoker

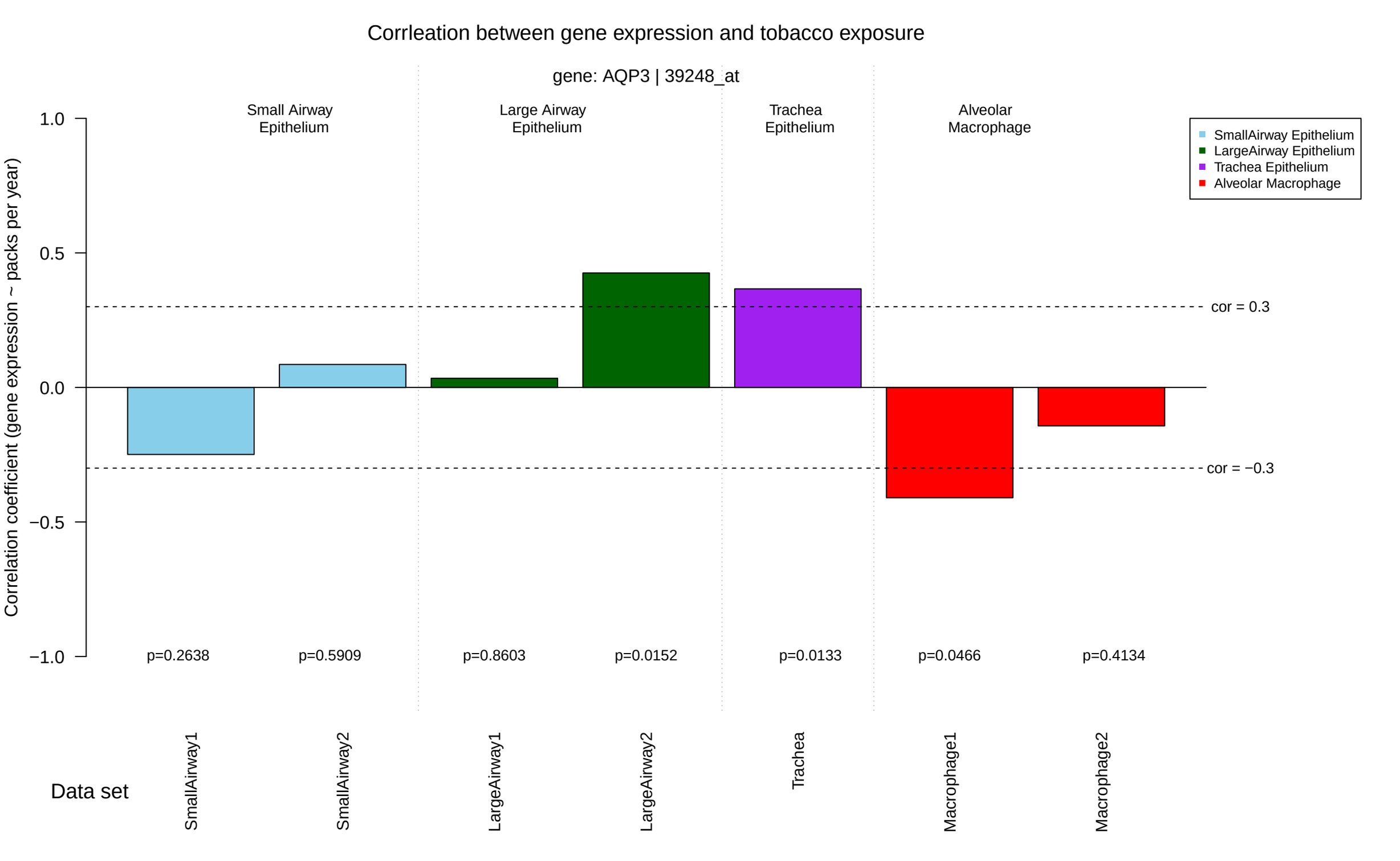

set: LargeAirway1 (p=0.0842)
AQP3 | 39248_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0018)
AQP3 | 39248_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=6e−04)
AQP3 | 39248_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.07)
AQP3 | 39248_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.443)
AQP3 | 39248_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AQP3 | 39248_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2638
p=0.5909
p=0.8603
p=0.0152
p=0.0133
p=0.0466
p=0.4134
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2